10.3 Hamilton cycles in the square of a graph

G^d

Given a graph G and a positive integer d, we denote by G^d the graph on $V(G)$ in which two vertices are adjacent if and only if they have distance at most d in G. Clearly, $G = G^1 \subseteq G^2 \subseteq \ldots$ Our goal in this section is to prove the following fundamental result:

Theorem 10.3.1. (Fleischner 1974)
If G is a 2-connected graph, then G^2 has a Hamilton cycle.

The proof of Theorem 10.3.1 will go roughly as follows. We start by finding a cycle C in G. Using induction, we shall cover the remaining vertices by C-paths in G^2. The first and last edges of those paths will be edges of G, like those of C. By deleting some of these edges and doubling others, we turn the union of C and all the C-paths into a multigraph with even degrees, and find an Euler tour in it. This Euler tour W will pass some vertices more than once, but all edges in such multiple passes will be edges of G. For all but one of the passes through a given vertex we can therefore try to replace its two G-edges by an edge of G^2 (Fig. 10.3.1), hoping to turn our Euler tour into a Hamilton cycle of G^2. The main difficulty will be to ensure that these lifts of passes are compatible, i.e., that we do not attempt to lift an edge at both its ends.

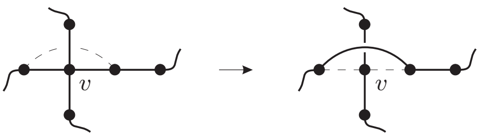

Fig. 10.3.1. Reducing the degree of v in W by lifting a pass

Lemma 10.3.2. *For every 2-connected graph G and $x \in V(G)$, there is a cycle $C \subseteq G$ that contains x as well as a vertex $y \neq x$ with $N_G(y) \subseteq V(C)$.*

Proof. If G has a Hamilton cycle, there is nothing more to show. If not, let $C' \subseteq G$ be any cycle containing x; such a cycle exists, since G is 2-connected. Let D be a component of $G - C'$. Assume that C' and D are chosen so that $|D|$ is minimum. Since G is 2-connected, D has at least two neighbours on C'. Then C' contains a path P between two such neighbours u and v, whose interior $\mathring{P}$ does not contain x and has no neighbour in D (Fig. 10.3.2).

Replacing P in C' by a u–v path through D, we obtain a cycle C that contains x and a vertex $y \in D$. If y had a neighbour z in $G - C$, then z would lie in a component $D' \subsetneq D$ of $G - C$, contradicting the choice of C' and D. Hence all the neighbours of y lie on C, and C satisfies the assertion of the lemma. $\square$

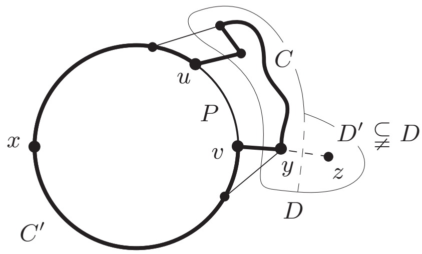

Fig. 10.3.2. The proof of Lemma 10.3.2

For our proof of Theorem 10.3.1 we need some more definitions. Let G be a multigraph, and W a walk in G. A *pass* of W *through* a vertex x is a subwalk of the form $uexfv$, where e and f are edges. (We also count $uexfv$ as a pass of W if $W = xfv \dots uex$.) By *lifting* this pass we mean replacing it in W by a new u–v edge if $u \neq v$, or by the single vertex u if $u = v$. A *multipath* is a multigraph obtained from a path by replacing some of its edges by double edges. Given $C \subseteq G$, we define a C*-trail* to be either a C-path or a cycle meeting C in exactly one vertex.

pass

lift

multipath

C-trail

(1.2.1)
(1.8.1)

Proof of Theorem 10.3.1. Let $G = (V, E)$ be a 2-connected graph. We prove the following stronger assertion by induction on $|G|$:

$G = (V, E)$

> *For every vertex $x \in V$ there is a Hamilton cycle in G^2 whose edges at x lie in E.*

x

If G is hamiltonian, there is nothing more to show. If not, let C and y be as provided by Lemma 10.3.2. For $i = 1, 2$ let $r_i, s_i \in V(C)$ and $g_i, h_i \in E(C)$ be such that

C, y

$$C = x g_1 r_1 \dots s_1 h_1 y h_2 s_2 \dots r_2 g_2 x\,;$$

$r_i, s_i;\ g_i, h_i$

see Figure 10.3.3. (These vertices and edges need not all be distinct.)

Our first aim is to construct for every component D of $G - C$ a set of C-trails in $G^2 + \overline{E}$, where $\overline{E}$ will be a set of additional edges parallel to edges of G. Every vertex of D will lie on exactly one such trail, and every edge of such a trail that is incident with a vertex of C will lie in E or in $\overline{E}$.

If D consists of a single vertex u, we pick any C-trail in G containing u, and let E_D be the set of its two edges. If $|D| > 1$, let $\tilde{D}$ be the (2-connected) graph obtained from G by contracting $G - D$ to a vertex $\tilde{x}$. Applying the induction hypothesis to $\tilde{D}$, we obtain a Hamilton cycle $\tilde{H}$ of $\tilde{D}^2$ whose edges at $\tilde{x}$ lie in $E(\tilde{D})$. Write $\tilde{E}$ for the set of those edges of $\tilde{H}$ that are not edges of G^2; these include its two edges at $\tilde{x}$. Replacing the edges from $\tilde{E}$ by edges of G or new edges $\overline{e} \in \overline{E}$, we shall turn $E(\tilde{H})$ into the edge set of a union of C-trails.

Consider an edge $uv \in \tilde{E}$, with $u \in D$. Then either $v = \tilde{x}$, or u and v have distance at most 2 in $\tilde{D}$ but not in G, and are hence neighbours of $\tilde{x}$ in $\tilde{D}$. In either case, G contains a u–C edge. Let E_D be obtained from $E(\tilde{H}) \setminus \tilde{E}$ by adding at every vertex $u \in D$ as many u–C edges from E as u has incident edges in $\tilde{E}$; if u has two incident edges in $\tilde{E}$ but in G sends only one edge e to C, we add both e and a new edge $\overline{e}$ parallel to e. Then every vertex of D has the same degree (two) in (V, E_D) as in $\tilde{H}$, so E_D is the edge set of a union of C-trails. Let

$\overline{e}$

G_0

$$G_0 := \Big(V, E(C) \cup \bigcup_D E_D\Big)$$

be the union of C and all these C-trails, for all components D of $G - C$ together.

Our next aim is to turn G_0 into an Eulerian multigraph by doubling some edges of C. Since G_0 is connected, it will suffice to do this in such a way that all degrees become even (Theorem 1.8.1).[2] The vertices of G_0 outside C already have degree 2. To make the degrees even also at the vertices of C we consider these in reverse order, starting with x and ending with r_1. Let u be the vertex currently considered, and let v be the vertex to be considered next. Add a new edge $\overline{e}$ parallel to $e = uv$ if and only if u has odd degree in the multigraph obtained from G_0 so far. When finally $u = r_1$ is considered, every other vertex has even degree, so r_1 must have even degree too (Proposition 1.2.1), and no edge parallel to g_1 will be added.

$\overline{e}$

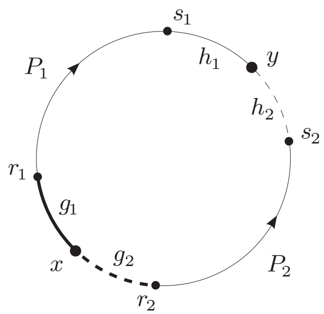

Fig. 10.3.3. Broken edges may not exist in G_1; bold edges are known to be single (if they exist)

From the Eulerian multigraph thus obtained we may now delete one or two double edges, as follows (Fig. 10.3.3). If g_2 has a parallel edge $\overline{g}_2$, we delete both g_2 and $\overline{g}_2$. If h_2 has a parallel edge $\overline{h}_2$, we delete h_2 and $\overline{h}_2$ unless this (together with the deletion of g_2 and $\overline{g}_2$) would

[2] To deduce the multigraph version of Theorem 1.8.1, subdivide every edge once to obtain a simple graph.

disconnect our multigraph. We write G_1 for the Eulerian multigraph thus obtained, $\overline{E} = E(G_1) \smallsetminus E(G^2)$ for the set of all its new parallel edges, and $C_1 := G_1[V(C)]$.

G_1 $\overline{E}$ C_1

Let us note two properties of G_1, which follow from its construction and the definition of y:

$$\textit{The edges of } G_1 \textit{ at vertices of } C \textit{ all lie in } E \cup \overline{E}. \tag{1}$$

$$N_{G_1}(y) \subseteq \{s_1, s_2\}\textit{; thus, } y \textit{ has degree 2 or 4 in } G_1. \tag{2}$$

Let $P_1 = x_0^1 \dots x_{\ell_1}^1$ be the (maximal) x–y multipath in C_1 containing g_1, and let $P_2 = x_0^2 \dots x_{\ell_2}^2$ be the multipath consisting of the other edges of C_1. Unless P_2 is empty, we think of it as running from $x_0^2 \in \{x, r_2\}$ to $x_{\ell_2}^2 \in \{y, s_2\}$. We write e_j^i for the x_{j-1}^i–x_j^i edge of P_i in $E(C)$, and $\overline{e}_j^i$ for its possible parallel edge in $\overline{E}$ $(i = 1, 2)$.

ℓ_i, x_j^i P_i $e_j^i, \overline{e}_j^i$

Our plan is to find an Euler tour W_1 of G_1 that can be transformed into a Hamilton cycle of G^2. In order to endow W_1 more easily with the required properties, we shall not define it directly. Instead, we shall derive W_1 from an Euler tour W_2 of a related multigraph G_2, which we define next.

For $i = 1, 2$ and every $j = 1, \dots, \ell_i - 1$ such that $\overline{e}_{j+1}^i \in G_1$, we delete e_j^i and $\overline{e}_{j+1}^i$ from G_1 and add a new edge f_j^i joining x_{j-1}^i to x_{j+1}^i; we shall say that f_j^i *represents* the path $x_{j-1}^i e_j^i x_j^i \overline{e}_{j+1}^i x_{j+1}^i \subseteq P_i$ (Fig. 10.3.4). Note that every such replacement leaves the current multigraph connected, and it preserves the parity of all degrees. Hence, the multigraph G_2 obtained from G_1 by all these replacements is Eulerian.

f_j^i G_2

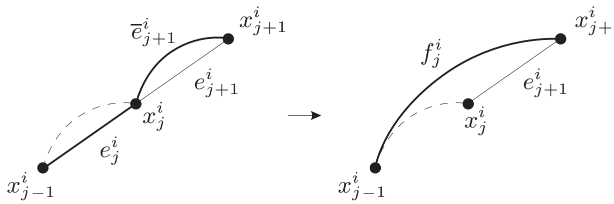

Fig. 10.3.4. Replacing e_j^i and $\overline{e}_{j+1}^i$ by a new edge f_j^i

Pick an Euler tour W_2 of G_2. To transform W_2 into an Euler tour W_1 of G_1, replace every edge in $E(W_2) \smallsetminus E(G_1)$ by the path it represents.

W_2 W_1

By (2), there are either one or two passes of W_1 through y. If $d(y) = 4$ then G_1 is connected but $G_1 - \{h_2, \overline{h}_2\}$ is not, by definition of G_1. Hence the proper y–y subwalk W of W_1 starting or ending with the edge h_2 must end or start with the edge $\overline{h}_2$: otherwise it would contain an s_2–y walk in $G_1 - \{h_2, \overline{h}_2\}$, and deleting h_2 and $\overline{h}_2$ from G_1 could not affect its connectedness. Therefore W_1 has no pass through y containing both h_2 and $\overline{h}_2$: this would close the subwalk W and thus imply $W = W_1$, contrary to the definition of W. Reversing W if necessary, we may thus assume:

If W_1 contains two passes through y, then one of these contains h_1 and h_2. (3)

Q

Our plan is to transform W_1 into a Hamilton cycle of G^2 by lifting, at every vertex $v \in C$, all but one of the passes of W_1 through v. We begin by marking the one pass at each vertex that we shall not lift. At x we mark an arbitrary pass Q of W_1. At y we mark the pass containing h_1. For every $v \in V(C) \setminus \{x, y\}$ there is a unique pair (i, j) such that $v = x^i_j$. If $j \geqslant 1$, we mark the pass through $v = x^i_j$ that contains e^i_j. If $j = 0$ (which can happen only if $v = x^2_0 = r_2$), we mark the pass through v that contains $\overline{e}^2_1$ if $\overline{e}^2_1 \in G_1$; otherwise we mark an arbitrary pass through v. At every vertex $v \notin C$ we mark the unique pass of W_1 through v. Thus:

At every vertex $v \in G$ we marked exactly one pass through v. (4)

To avoid conflicts when we later lift the unmarked passes, we need that no edge of W_1 is left unmarked at both its ends:

For every edge $e = uv$ in W_1 we marked at least one of the two passes of W_1 containing e (one through u, the other through v). If $u = x$, we marked the pass through v. (5)

This is clear for edges not in C_1. For every edge $e \in C_1$, there is a unique pair (i, j) such that $e = e^i_j$ or $e = \overline{e}^i_j$; then $j \geqslant 1$. If $e = e^i_j$, we marked the pass of W_1 through x^i_j that contains e; for $e = h_2$ this follows from (3). If $e = \overline{e}^i_j$, we marked the pass through x^i_{j-1} containing e. Indeed, note first that e is not incident with x: recall that $\overline{g}_1$ never existed, and if $\overline{g}_2$ existed it was deleted in the definition of G_1. Hence unless P_2 starts at r_2 and $e = \overline{e}^2_1$, an edge f^i_{j-1} was defined to represent the path $x^i_{j-2} e^i_{j-1} x^i_{j-1} \overline{e}^i_j x^i_j$. Since W_2 contained f^i_{j-1}, this path is a pass in W_1. We marked this pass, because it is a pass through x^i_{j-1} containing e^i_{j-1}. Finally, if P_2 starts at r_2 and $e = \overline{e}^2_1$, we marked the pass through r_2 containing e explicitly. This completes the proof of (5).

By (1), all unmarked passes lift to edges of G^2. As different unmarked passes never share an edge (5), lifting them all at once turns W_1 into one closed walk $\overline{H}$ in $G^2 + \overline{E}$ (which inherits the cyclic ordering of its edges from W_1). By (4), $\overline{H}$ still contains all the vertices v of G: if uv is an edge of a pass marked at v, then $\overline{H}$ contains either the edge uv or the lift wv of a pass $weufv$. Also by (4), $\overline{H}$ traverses every vertex only once. In particular, $\overline{H}$ cannot contain a pair of parallel edges. We can therefore replace every edge $\overline{e}$ in $\overline{H}$ by its parallel edge $e \in E$, to obtain a Hamilton cycle H of G^2. Since we marked Q, and by (5) no edge of Q was lifted at its other end, $\overline{H}$ contains the edges of Q. By (1), these lie in $E \cup \overline{E}$. Hence the edges of H at x lie in E, as desired. □

Fleischner's theorem has a natural extension to infinite graphs, which is much harder to prove:

Theorem 10.3.3. (Georgakopoulos 2009)
The square of every 2-connected locally finite graph contains a Hamilton circle.

We close the chapter with a far-reaching conjecture generalizing Dirac's theorem:

Conjecture. (Seymour 1974)
Let G be a graph of order $n \geqslant 3$, and let k be a positive integer. If G has minimum degree

$$\delta(G) \geqslant \frac{k}{k+1}\, n\,,$$

then G has a Hamilton cycle H such that $H^k \subseteq G$.

For $k = 1$, this is precisely Dirac's theorem. The conjecture was proved for large enough n (depending on k) by Komlós, Sárközy and Szemerédi (1998).

Exercises

1. An oriented complete graph is called a *tournament*. Show that every tournament contains a (directed) Hamilton path.

2. Show that every uniquely 3-edge-colourable cubic graph is hamiltonian. ('Unique' means that all 3-edge-colourings induce the same edge partition.)

3. Given an even positive integer k, construct for every $n \geqslant k$ a k-regular graph of order $2n+1$.

4. Prove or disprove the following strengthening of Proposition 10.1.2: 'Every k-connected graph G with $|G| \geqslant 3$ and $\chi(G) \geqslant |G|/k$ has a Hamilton cycle.'

5. Let G be a graph, and $H := L(G)$ its line graph.
   (i) Show that H is hamiltonian if G has a spanning Eulerian subgraph.
   (ii)$^+$ Deduce that H is hamiltonian if G is 4-edge-connected.

6. (i)$^-$ Show that hamiltonian graphs are 1-tough.
   (ii) Find a graph that is 1-tough but not hamiltonian.

7. Prove the toughness conjecture for planar graphs. Does it hold with $t = 2$, or even with some $t < 2$?